MULT20015 Elements of Quantum Computing
Assignment 2

Due: 10th October 2024 5pm (AEST)
Worth: Total points: 30. Weight for subject mark: 30%.
Number of problems: 5.

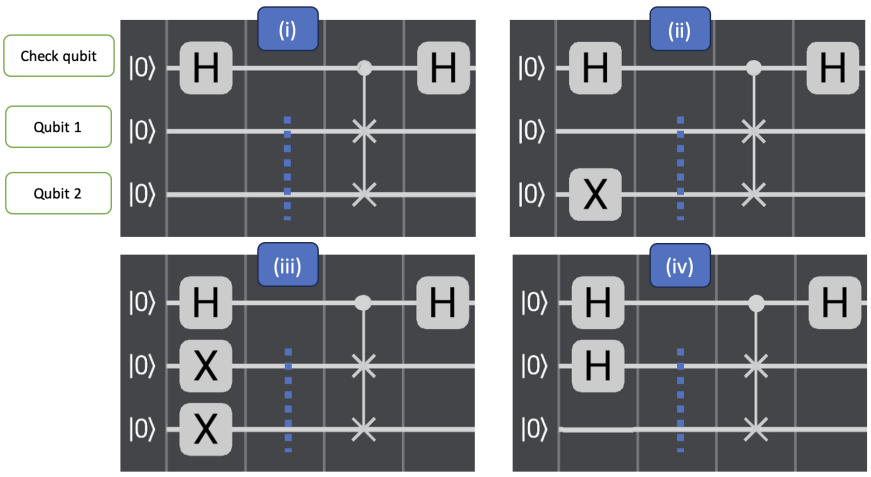
Submit your completed written work electronically as a PDF (no other formats accepted) to Canvas/LMS. You are welcome to use an app like `CamScanner` to convert your handwritten notes to pdf, and keep submissions under 10MB. Please put your name and student number on the front of your submission. Show all working.

While we encourage you to discuss this assignment with your classmates (*e.g.*, you can organize informal discussion session or meetups), every student must write their own solution based on their personal understanding. Copying full or partial solutions from other students may invalidate your submission, please remind yourself of UniMelb's Academic Integrity Policy and Penalties.

1 Algorithmic Primitives [0.5+0.5+0.5+1.0+1.0+1.0 = 4.5 points]

Recall that a SWAP gate is a two qubit gate that swaps the state of a two bit register: SWAP $|\psi\rangle \otimes |\phi\rangle = |\phi\rangle \otimes |\psi\rangle$. The controlled SWAP is an important three qubit gate. If the first qubit is zero then it does nothing (or identity) on the bottom two qubits. If the first qubit is one then it SWAPs the bottom two qubits. Of course the interesting thing happens when we send a superposition of one and zero through the top wire.

(a) Construct the following circuits using the QUI.



Using the QUI compute the probability of measuring "0" on the top wire (the "check qubit") for circuits (i), (ii), (iii), and (iv).

(b) The states on the bottom two wires at the blue dashed line (before the controlled SWAP) are: (i) $|0\rangle, |0\rangle$; (ii) $|0\rangle, |1\rangle$; (iii) $|1\rangle, |1\rangle$; (iv) $|+\rangle, |0\rangle$. For each of these states compute $|\langle \text{state of qubit 1} | \text{state of qubit 2}\rangle|^2$, e.g. for (i) $|\langle 0|0\rangle|^2$.

(c) Use your results from part (b) and compute $\Pr(0) = \frac{1}{2}(1 + |\langle \text{state of qubit 1} | \text{state of qubit 2}\rangle|^2)$ for each circuit. This probability should be equal to the probability for measuring "0" on the "check qubit" from the QUI in part (a).

(d) Let's consider two general states on the bottom wires $|\psi\rangle = a_0 |0\rangle + a_1 |1\rangle$ and $|\phi\rangle = b_0 |0\rangle + b_1 |1\rangle$. So the state of the register before the controlled SWAP is

$$|+\rangle \otimes |\psi\rangle \otimes |\phi\rangle \ .$$

Expand the plus state to $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$ and then apply the controlled SWAP

$$\text{cSWAP} = |0\rangle\langle 0| \otimes I \otimes I + |1\rangle\langle 1| \otimes \text{SWAP}$$

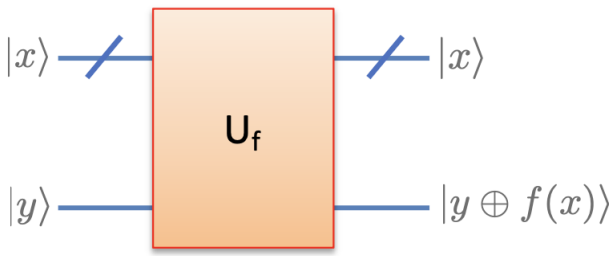
to the inital state, that is compute $|\Psi\rangle = \text{cSWAP}\, |+\rangle \otimes |\psi\rangle \otimes |\phi\rangle$. [Hint: don't use a matrix representation!]

(e) Now multiply the resulting state by the H gate on the top wire, i.e. $H \otimes I \otimes I\,|\Psi\rangle$. Let's call the resulting state $|\Psi'\rangle$. Write $|\Psi'\rangle$ resulting state as $|\Psi'\rangle = |0\rangle \otimes (\text{something}) + |1\rangle \otimes (\text{something else})$. Where "something" and "something else" are linear combinations of quantum states.

(f) The part of the state that represents measuring zero on the first qubit is $|0\rangle \otimes (\text{something})$. By only considering that part show that $\Pr(0) = \frac{1}{2}(1 + |\langle\psi|\phi\rangle|^2)$. [**Hints: (1)** Revise the lecture on multiqubit measurement. **(2)** Recall (number 1) $\otimes$ (number 2) = (number 1) $\times$ (number 2). **(3)** Recall that $\langle\psi|\phi\rangle \times \langle\phi|\psi\rangle = |\langle\psi|\phi\rangle|^2$ and $\langle\phi|\psi\rangle \times \langle\psi|\phi\rangle = |\langle\psi|\phi\rangle|^2$.]

2 Deutsch-Josza & Simon's [2.0+2.0+3.0 = 7.0 points]

Consider the Deutsch-Josza algorithm for the case where the function $f(x)$ is defined over n-bit numbers, x. In Lectures, the general circuit for function evaluation was given by:



Consider a set of functions f_i defined over four qubit ($n = 4$) inputs as follows:

$$\begin{aligned} f_1(x) &= 0 \\ f_2(x) &= 1 \\ f_3(x) &= x \mod 2 \\ f_4(x) &= (x+1) \mod 2\,. \end{aligned}$$

The functions above are either constant or balanced. If you are given an unknown function f_i where i is one of the 4 values from the set $\{1, 2, 3, 4\}$, your job is to develop a classical and a quantum algorithm, which can determine whether the function is constant or balanced.

(a) Describe a classical algorithm to solve the above problem. What is the elementary step? What is the cost function (under worst-case scenario) as a function of input size n? Find the Big-$\mathcal{O}$ class from the cost function.

(b) Use the Deutsch-Josza quantum algorithm to solve the above problem for $n = 4$. Determine the quantum circuit for each f_i and explain carefully how they work, that is step through the logic of this circuit. Provide a screen shot of each circuit from the QUI.

(c) Consider a related problem where we have a set of n functions defined over n-bit numbers x as:

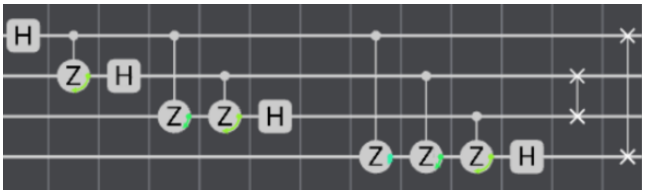
$$f_i(x) = x_i,$$

where x_i is the i'th bit of x ($i = 0..3$, counting from the least significant bit ($i = 0$)). Assume you are given the oracle corresponding to the function f_i but do not know its details. Devise a classical and quantum algorithm to determine the value of i. (Hint. For the quantum case, examine the upper register of the Deutsch-Josza algorithm) What is the elementary step in your classical and quantum implementations? Determine the worst-case number of elementary steps required for $n = 4$ for both the classical and quantum algorithms.

3 Quantum Phase Estimation [1.0+3.0+2.0+3.0 = 9.0 points]

(a) On the QUI, program a Y-rotation gate with a rotation angle of $\frac{2\pi}{3}$ and a global phase of $\frac{\pi}{3}$ directly on a $|\psi\rangle = (|0\rangle - \mathrm{i}\,|1\rangle)/\sqrt{2}$ state. What is the resulting state (including phase) when this rotation is applied to the $|\psi\rangle$ state? Show that $|\psi\rangle$ is an eigenstate of this rotation. Include your circuit in your answer.

(b) Construct the Quantum Phase Estimation (QPE) circuit described in the lectures to estimate the eigenvalue of the eigenstate, $|\psi\rangle$ for the Y-rotation gate specified in part (a). Use four qubits in the control (upper) register. You may make use of the following (inverse) quantum Fourier Transform for four qubits:



The angles of the controlled-Z rotations shown (in order, from left to right) are:

i $-\frac{\pi}{2}$ (with a global phase of $-\frac{\pi}{4}$)
ii $-\frac{\pi}{4}$ (with a global phase of $-\frac{\pi}{8}$)
iii $-\frac{\pi}{2}$ (with a global phase of $-\frac{\pi}{4}$)
iv $-\frac{\pi}{8}$ (with a global phase of $-\frac{\pi}{16}$)
v $-\frac{\pi}{4}$ (with a global phase of $-\frac{\pi}{8}$)
vi $-\frac{\pi}{2}$ (with a global phase of $-\frac{\pi}{4}$)

Include a screenshot of your circuit in your answer.

(c) Recall that if

$$U|\psi\rangle = \exp(2\,\mathrm{i}\,\pi\theta)\,|\psi\rangle\ ,$$

then QPE allows us to estimate θ using a quantum computer. Determine (using QUI) the measurement probabilities on the control/upper register of the QPE circuit in part (b). Use these measurements to estimate θ, and briefly compare you estimates of θ with your answer in part (a). Why are they not identical?

(d) Now consider the rotation $U = R_Y(2\pi/5)$ (with a global phase of $\pi/5$). Repeat (b)-(c) for this new rotation operation.

4 RSA & Shor's algorithm [1.0+(1.0+1.0) = 3.0 points]

(a) Alice chooses the two primes to construct her RSA keys:

$$p = 28813096529271137159477 11,\ \text{and}$$
$$q = 14323296331779246294235011689\,.$$

Determine the smallest valid RSA public key and its corresponding private key for Alice. Show your working with an explanation justifying your answer. You may use http://magma.maths.usyd.edu.au/calc/ .

(b) An experiment attempts to factor 511 using Shor's algorithm on a quantum computer. The experimenters choose a base of $a = 3$, and attempt to find the period, r, such that $a^r = 1 \mod 511$. After performing Shor's algorithm, the upper control register, containing $n = 18$ qubits is measured and the value $m = 100101010101010101_2$ is obtained.

i Based on this measured value, use the continued fraction expansion to find candidate periods, r. For each candidate period, r, test to determine if $a^r = 1 \mod 511$ or not. Show your working.

ii Use the period, r, obtained in part (i) to find the factors of 511. Show your working.

5 Grover [1.0 + 1.0 + 1.0 + 1.0 + 1.0 + (1.0 + 0.5) = 6.5 points]

In lectures and tutorials you have seen how to implement Grover's algorithm to find a particular number that you already know when constructing the oracle. In this problem we will construct an oracle that uses the properties of a solution to solve a problem where we don't know the exact solution in advance.

A quantum-sudoku is a 2×2 grid, filled with the numbers 0 and 1. The entries within each row and column must be unique, i.e. a row cannot contain the sequence 00 or 11. In this problem we label each entry as in Fig. a below;

v_0	v_1
v_2	v_3

(Fig. a)

1	0
0	1

(Fig. b)

(a) Write down four relations between the v_is that are equivalent to the uniqueness conditions on each row and column.

(b) Write down a quantum circuit that takes two input qubits corresponding to v_i and v_j, and sets an output ancilla qubit to 0 if they are the same and 1 if they are different.

(c) Write an oracle for Grover's algorithm that uses 4 qubits to represent the state of the sudoku board, 4 ancillas to hold the results of the uniqueness checks, and one output qubit initially prepared in the $|-\rangle$ state. The oracle should mark valid states with a phase of -1. Program this into QUI and include a screenshot.

Note: you should do some uncomputation to return the ancilla qubits to the $|0\rangle$ state at the end of the oracle.

(d) Program the mean inversion step into QUI and run Grover's algorithm. How many iterations do you need to have the maximum probability of finding a valid solution to the sudoku? Include a link to the QUI circuit.

Note: You can select multiple gates in QUI with the pointer, and copy and paste to repeat an iteration of Grover's algorithm.

(e) In a "real" sudoku, some squares on the grid are already filled. How can we program this additional constraint into our oracle for Grover's algorithm? Consider the specific case where $v_0 = 1$. How many Grover iterations are optimal in this case? Show your circuit and the corresponding QUI output.?

(f) i. How does the computational complexity of this problem scale with n, the length of a row/column? Assume that the ratio between the number of possible grids and the number of possible solutions to the sudoku is $f(n)$[1]. Give your answer in big-$\mathcal{O}$ notation.

ii. A full 3×3 sudoku also has the condition of each subsquare containing every number exactly once. If you created an oracle including this condition, what is the computational complexity of the algorithm in big-$\mathcal{O}$ notation?

[1] It turns out the mathematics of sudoku is quite complicated